

Emotion Classification in Text using TF-IDF, LSTM and BERT

Author: Mateusz Kantorski

1. Introduction

Text-based emotion recognition is an important task in natural language processing (NLP), with applications in sentiment analysis, social media monitoring, and human-computer interaction. The goal of this project is to classify short English sentences into six emotions: sadness, joy, love, anger, fear, and surprise. Three different approaches are evaluated: a baseline Logistic Regression using TF-IDF features, a Long Short-Term Memory (LSTM) network, and a pretrained Transformer-based model (BERT).

The motivation for this task is to explore how classical NLP methods compare with deep learning architectures for a multi-class emotion classification problem.

2. Dataset

The Emotion dataset is employed, containing short English sentences labeled with six emotions: sadness, joy, love, anger, fear, and surprise.

- **Size:** approximately 20,000 samples
- **Split:** 80% training, 10% validation, 10% test
- **Source:** <https://huggingface.co/datasets/emotion>
The dataset is well-balanced for the major emotions, while the **love** and **surprise** classes have fewer samples

3. Methods

3.1 Baseline: TF-IDF + Logistic Regression

- Texts are tokenized by splitting on spaces and lowercased.
- TF-IDF features are computed using unigrams and bigrams, limited to 20,000 features.
- Logistic Regression is trained using a multinomial (softmax) approach on the TF-IDF features.

3.2 LSTM

- Each word is converted to an integer index using a vocabulary of 20,000 most frequent words.
- Sequences are padded to length 50.

- Embedding dimension: 128, hidden size: 128, dropout: 0.3
- CrossEntropyLoss and Adam optimizer with learning rate 0.001
- Training: 15 epochs, batch size 64, gradient clipping 1.0

3.3 BERT (distilbert-base-uncased)

- HuggingFace **distilbert-base-uncased** is fine-tuned for 4 epochs on the training set.
- AdamW optimizer, batch size 16, learning rate 2e-5
- Validation is monitored at the end of each epoch to prevent overfitting.
- Tokenizer and model weights are saved for inference

4. Training Procedure

The dataset is split into training, validation, and test sets as described in Section 2.

- **TF-IDF + Logistic Regression:** The model is trained on the training set and evaluated on the test set. Macro F1, accuracy, and the classification report are computed to assess performance. No separate validation set is used.
- **LSTM:** The model is trained on the training set for 15 epochs with batch size 64. After each epoch, the model is evaluated on the validation set to monitor learning progress and prevent overfitting. Gradient clipping is applied. Final metrics, including macro F1, accuracy, and the classification report, are computed on the test set.
- **BERT:** The pretrained **cdistilbert-base-uncased** model is fine-tuned on the training set for 4 epochs with batch size 16 and a learning rate of 2e-5. Validation performance is monitored at the end of each epoch. Final metrics, including macro F1, accuracy, and the classification report, are computed on the test set. Additionally, a confusion matrix is generated to visualize per-class performance.

5. Results

5.1 Accuracy and Macro F1

The overall performance of the three models on the test set is summarized below. Both accuracy and macro F1 are reported to provide a balanced evaluation across all classes.

Model	Accuracy	Macro F1
TF-IDF + Logistic Regression	0.8845	0.8248
LSTM	0.8500	0.8000
DistilBERT	0.9430	0.9214

Observation: DistilBERT achieves the highest overall performance due to its contextualized embeddings. LSTM benefits from sequential modeling, whereas TF-IDF + Logistic Regression provides a surprisingly strong baseline despite its simplicity.

5.2 Classification Reports

5.2.1 TF-IDF + Logistic Regression

Emotion	Precision	Recall	F1-score	Support
sadness	0.91	0.95	0.93	581
joy	0.86	0.96	0.91	695
love	0.83	0.62	0.71	159
anger	0.90	0.83	0.87	275
fear	0.90	0.84	0.87	224
surprise	0.94	0.52	0.67	66

- **Accuracy:** 0.8845
- **Macro F1:** 0.8248

Observation: TF-IDF achieves high precision and recall for major classes (sadness, joy, anger, fear) but underperforms on smaller classes (love and surprise).

5.2.2 LSTM

Emotion	Precision	Recall	F1-score	Support
sadness	0.96	0.88	0.92	581
joy	0.83	0.92	0.87	695
love	0.58	0.70	0.64	159
anger	0.90	0.77	0.83	275
fear	0.87	0.85	0.86	224
surprise	0.76	0.59	0.67	66

- **Accuracy:** 0.8500
- **Macro F1:** 0.8000

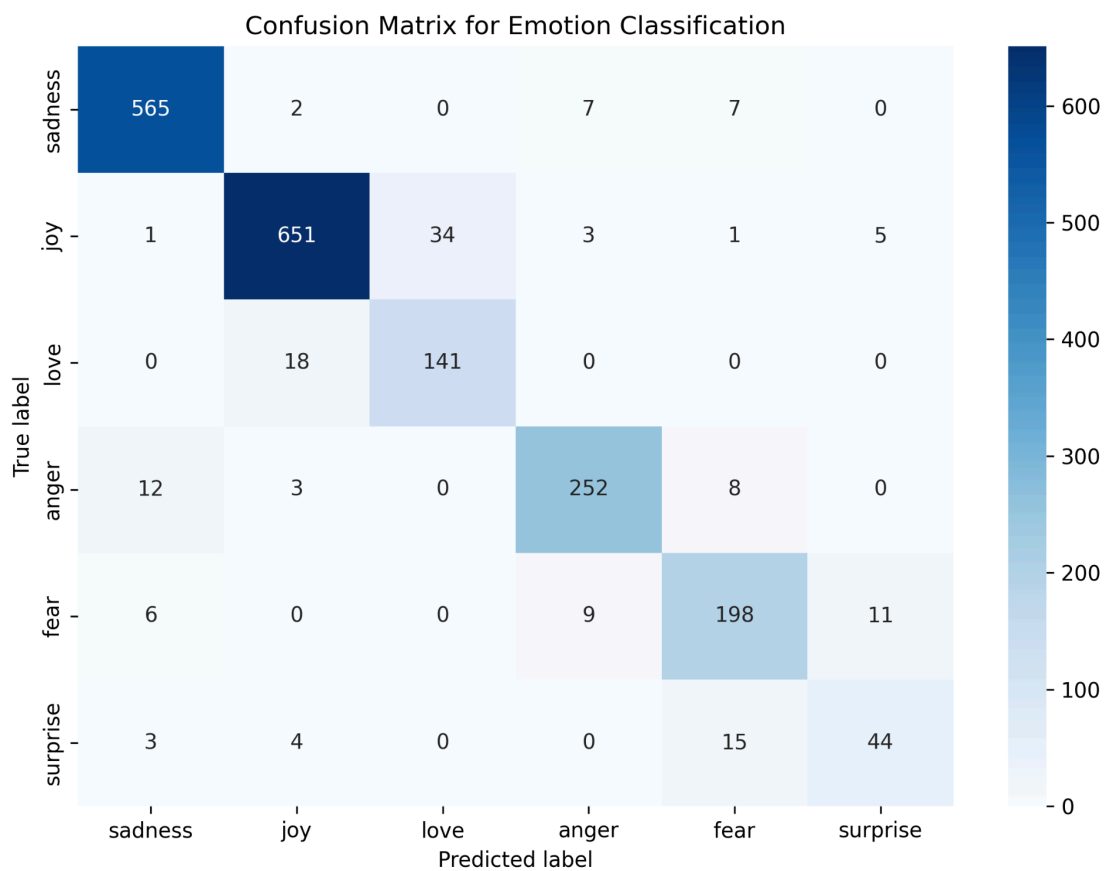
Observation: LSTM benefits from modeling sequential dependencies in the text, improving recall for smaller classes compared to TF-IDF. However, it still struggles in classes with very few examples (love and surprise).

5.2.3 DistilBERT

Emotion	Precision	Recall	F1-score	Support
sadness	0.96	0.97	0.97	581
joy	0.96	0.94	0.95	695
love	0.81	0.89	0.84	159
anger	0.93	0.92	0.92	275
fear	0.86	0.88	0.87	224
surprise	0.73	0.67	0.70	66

- **Accuracy:** 0.9430
- **Macro F1:** 0.9214

Observation: DistilBERT achieves the best performance across all classes, effectively capturing contextual information and handling nuanced expressions. The main confusion occurs between love and joy, which share similar vocabulary, but overall the model outperforms both LSTM and TF-IDF.



6. Discussion

- **TF-IDF + Logistic Regression:** Quick and interpretable, but does not capture word order or context.
- **LSTM:** Models sequence information, can improve with pre-trained embeddings (GloVe/FastText).
- **BERT:** Contextual embeddings significantly improve performance, making it the best model overall.

Limitations:

- Small classes (love, surprise) remain challenging.
- LSTM may require more epochs or larger embedding sizes to match BERT.

7. Conclusion

This project demonstrates that text-based emotion classification can be tackled with both classical and deep learning methods. While TF-IDF + Logistic Regression provides a strong baseline, BERT outperforms other approaches due to its contextual understanding of language. LSTM performs moderately well and can be improved with pretrained embeddings.

Future work could explore data augmentation, multi-lingual datasets, or more advanced transformer models for improved performance.

8. References

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